Sawyer Greaves

greaves.sawyer@gmail.com | Website

EDUCATION

M.S. Mechanical Engineering May 2027
University of Utah **3.829 GPA**

B.S. Mechanical Engineering Dec. 2025
University of Utah **3.960 GPA**

TECHNICAL SKILLS

ROS2, Python, C, embedded systems, MATLAB, manufacturing, simulation, robot arms, SOLIDWORKS, OnShape, controls, OOP.

EXPERIENCE

Research Assistant May 2025 - Present
Magnetic & Medical Robotics Lab, Salt Lake City, UT

- See *Publications* for details.

Research Engineering Intern Aug. 2023 – Aug. 2024
Sublime Laser, Salt Lake City, UT

- Developed femto-second laser machining process for guidewires, catheters, and stents.

PUBLICATIONS

S.R. Greaves, C.R. Lambert, K. Dhekar, J.J. Abbott. Academy Transaction Note: Capturing Objects in Microgravity Using Induction. *Acta Astronautica*. **Under revision for resubmission**.

- Explores a concept for a device to capture conductive objects without knowledge of object properties or position. Corresponding author.

L.M. Barker, T.J. Allen, **S.R. Greaves**, J.J. Abbott. Force and Torque Induced on a Conductive Nonmagnetic Cylinder by a Rotating Magnetic Dipole. *IEEE Transactions on Magnetics*. **Under review**.

- Models the force and torque on conductive cylinder. My contributions expanded this data set to cylinders of all shapes and sizes.

S.R. Greaves, J.J. Abbott. Interpolating Models of Magnetically Induced Wrenches in SO(3). *IEEE Robotics and Automation Letters*. **In preparation**.

- Explores a novel method for interpolating a sparse wrench data set based on the orientation of the object by treating orientation quaternions as basis vectors. Corresponding author.

T.J. Allen, K.A. Ben-Jacob, R.R. Scoggan, T.C. Wilcox, **S.R. Greaves**, E.M. VanderHoeven, A.P. Newey, L.E. Martin, D.A. Wall, G.F. Tabor, J.J. Abbott. First Demonstration of Six-Degree-of-Freedom Manipulation Using Magnetic Induction Enabled by a Novel Neutral-Buoyancy Simulation of Microgravity. *IEEE Transactions on Robotics*. **In preparation**.

- Explores a new simulation platform for full 6DOF magnetic manipulation of conductive objects. My contributions allowed for online vision-based state estimation.